

Ideation Phase

Empathize & Discover

Date	Project Name	Maximum Marks
13 July 2026	India's Agricultural Crop Production Analysis using Tableau	4 Marks

Empathy Map Canvas:

Problem Statement	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	An agricultural analyst and policy researcher specializing in national food security and regional crop yields.	Evaluate historical data across states and seasons to identify production trends and optimize resource distribution.	I am buried in massive, fragmented spreadsheets that make cross-regional comparisons slow and error-prone.	Traditional spreadsheet tools lack the multi-dimensional processing power and dynamic maps needed to track complex geospatial patterns.	Frustrated and overwhelmed by the time wasted cleaning raw rows instead of deriving actionable strategic insights.
PS-2	A regional agricultural officer and data-driven decision maker advising local farming communities.	Identify the most productive, high-yielding crop varieties suited for specific seasonal cycles and soil zones.	It is incredibly difficult to rapidly uncover hidden correlations or compare yield metrics against regional baselines.	Static charts and tabular data formats fail to offer intuitive filtering or real-time visual comparisons.	Disadvantaged and disconnected from the critical insights needed to confidently guide modern farming strategies.

1. SAYS

- "I need to look at clear historical crop statistics over multiple years rather than scrolling through rows of spreadsheets."
- "It is very time-consuming to manually isolate seasonal trends like Kharif or Rabi cycles across different states."
- "I want to easily see which crops perform best in specific high-yielding zones without calculating it manually."
- "Our teams require visual reports and geographic map views that quickly pinpoint falling crop yields."
- "It takes way too long to find patterns when our raw datasets are split across various regional documents."

2. THINKS

- Which states show the highest total agricultural production over the last two decades?
- Is there a clean way to track how cultivated area changes affect final crop yield?
- How can I present these complex seasonal cycles clearly to regional officers without causing confusion?
- Can visual dashboard filters help our researchers query data without needing direct SQL skills?
- How do environmental shifts or shifting crop choices impact regional food security targets?

3. DOES

- Compiles massive datasets covering Indian crop cycles from 1997 through 2021.
- Compares regional farming statistics across different states to check historical variations.
- Builds manual presentation charts and summary tables for regional agricultural planning meetings.
- Uses calculated parameters to extract productivity and crop yield indicators from raw area statistics.
- Filters records continuously to separate distinct Kharif, Rabi, and Zaid seasonal results.

4. FEELS

- Slowed down by large, unorganized data tables that lag during basic analysis.
- Uncertain when working with static files that lack geographical context or map integration.
- Eager to discover long-term trends and hidden production drops across minor agricultural sectors.
- Relieved when complex historical parameters are brought together into a single visual view.
- Motivated to deliver clearer field guidance to farmers using concrete data insights.

5. PAINS

- Processing complex agricultural datasets often leads to severe performance slowdowns in traditional tools.
- Static spreadsheets completely hide underlying geographic correlations and regional production trends.
- Cross-referencing production records across 24 separate years involves heavy manual sorting.
- Calculating specific crop yields across distinct states requires writing redundant formulas manually.
- The complete lack of responsive visual elements makes it difficult to present insights to non-technical partners.
- Isolating minor seasonal variations across localized crop types becomes highly error-prone.
- Fragmented layouts complicate the baseline tracking needed for long-term food security planning.

6. GAINS

- Clean, interactive visualization dashboards that present huge datasets clearly on a single page.
- Significant reduction in manual data analysis effort through interactive, user-driven dashboard filters.
- Immediate discovery of high-performing crop varieties via clean geo-spatial map views.

- Seamless tracking of long-term production metrics across precise timelines.
- Faster turnaround times when creating performance reports for policy makers and agricultural advisors.
- Accurate calculation of regional productivity using automated calculated fields.
- Improved strategic planning capabilities that directly support local farming communities.